

SHAHIL ABBASS

Data Scientist

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PROFESSIONAL SUMMARY

Aspiring Data Scientist with strong foundation in Machine Learning, Deep Learning, Computer Vision, and NLP. Skilled in Python, TensorFlow, Scikit-learn, Pandas, and FastAPI for building AI-powered applications and predictive models. Developed real-time intelligent systems including AI Voice Assistants, sentiment analysis tools, and computer vision applications. Passionate about transforming data into actionable insights and deploying scalable AI solutions.

TECHNICAL SKILLS

- **Programming Languages:** Python
- **Data Analysis & Visualization :** Numpy, Pandas, Matplotlib, Seaborn, Tableau, Power BI
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, Keras, PyTorch
- **Computer Vision & NLP:** OpenCV, NLTK, Hugging Face, LangChain, YOLO
- **Databases:** MySQL, MongoDB
- **Tools & IDEs:** Jupyter Notebook, Google Colab, VS Code, PyCharm

PROJECTS

AETHER BI assistant | github.com/Shahil-Abbas/AETHER-BI-assistant

Tech: Python, Flask, SQLite, Pandas, JavaScript, Tailwind CSS, Plotly, Ollama, Llama 3.2

- Developed a full-stack **Generative BI (Business Intelligence) Assistant** using Flask, SQLite, JavaScript, and Local LLMs.
- Built an AI-powered **Natural Language to SQL Engine** using Ollama (Llama 3.2) for automated business analytics.
- Implemented dynamic **data visualization system** with Plotly supporting bar, line, pie, scatter, and heatmap charts.
- Designed an interactive **analytics dashboard** with KPI cards, forecasting engine, SQL editor, and AI chat assistant.
- Added support for **CSV and Excel dataset uploads** with automatic schema detection and SQLite integration.
- Engineered a secure SQL execution pipeline with **SQL validation, syntax checking, and query safety filtering**.
- Developed real-time **AI insight streaming system** using Server-Sent Events (SSE) for business recommendations.
- Created predictive analytics module with **trend forecasting and linear regression-based projections**.
- Implemented responsive premium UI using **Tailwind CSS, Glassmorphism, Plotly, and JavaScript**.

AI voice assistant using Python and NLP | github.com/Shahil-Abbas/Ai_assistant

Tech: Python, NLP, SpeechRecognition, pytsx3, OpenCV, PyAudio, NumPy, Automation

- Developed a voice-controlled AI assistant using Python, NLP, and speech recognition techniques.
- Implemented real-time voice command processing with automated task execution and contextual response handling.
- Integrated text-to-speech and speech-to-text modules for conversational interaction.
- Built intelligent mathematical expression handling and dynamic command execution system.
- Designed modular architecture for scalability and future AI feature integration.

Gesture Control PC | github.com/Shahil-Abbas/gesture_control_pc

Tech: Python, OpenCV, MediaPipe, NumPy, PyAutoGUI, Computer Vision, Hand Tracking

- Built a real-time hand gesture recognition system for controlling PC operations using Computer Vision.
- Implemented finger tracking and gesture detection using OpenCV and MediaPipe.
- Developed virtual mouse functionalities including cursor movement, clicking, and scrolling through gestures.
- Optimized real-time webcam processing to improve gesture detection responsiveness and interaction accuracy.
- Designed modular Python architecture for scalable gesture-based automation applications.

INTERNSHIP EXPERIENCE

Data Scientist | SMEC Technologies

- Assisted in data preprocessing, cleaning, and exploratory data analysis (EDA) using Python libraries such as Pandas, NumPy, and Matplotlib to identify trends and patterns in datasets
- Developed machine learning models for classification and prediction tasks using Scikit-learn and evaluated model performance using accuracy, precision, recall, and F1-score metrics
- Built interactive data visualizations and dashboards to present analytical insights and support data-driven decision making processes
- Performed feature engineering, data transformation, and model optimization techniques to improve machine learning model performance and reduce overfitting
- Collaborated with team members on AI and data science projects involving NLP, computer vision, and predictive analytics using Python and deep learning frameworks
- Built classification models using Logistic Regression, Random Forest, and Decision Trees for predictive analysis tasks

EDUCATION

Diploma In Computer Engineering | Govt Polytechnic College Kasaragod | 2022–2025

CGPA:8.15

LANGUAGES

English | Malayalam